

19.11.16(E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
Nov – Dec 2016

Max. Marks: 100

Class: SYBTECH

Name of the Course: Database Management Systems

Duration: 03 hours

Semester: III

Branch: IT

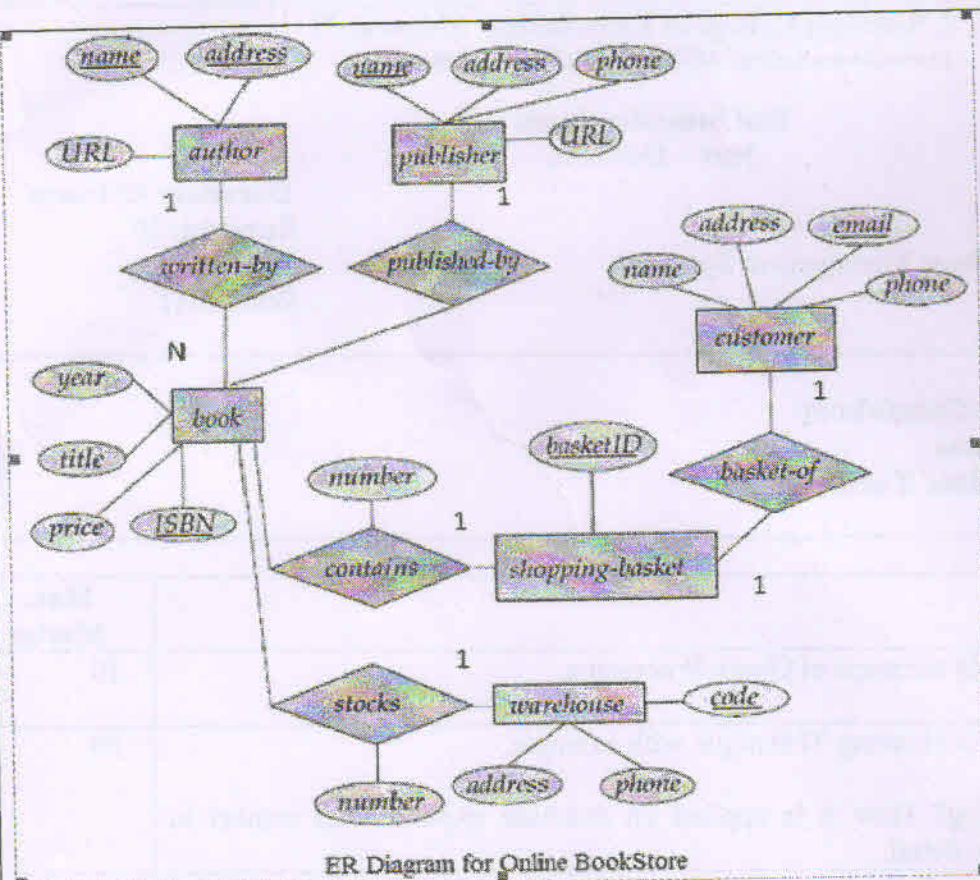
Course Code: UITC305

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q1 (a)	Explain in details the steps of Query Processing.	10
Q1 (b)	Explain the Static Hashing Technique with example. OR What is Indexing? How it is applied on database explain with respect to sparse indices in detail.	10
Q2 (a)	What are Triggers? Explain with example creating, inserting and updating Triggers.	10
Q2 (b)	Explain 1NF, 2 NF, 3 NF and BCNF with example.	10
Q 3 (a)	A university registrar's office maintains data about the following entities: 1. Courses, including number, title, credits, syllabus, and prerequisites; 2. Course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom; 3. Students, including student-id, name, and program; 4. Instructors, including identification number, name, department, and title. Further, the enrolment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modelled. Construct an EER diagram for the registrar's office. Document all assumptions that you make about the mapping constraints. OR Convert following EER diagram in to Relational Model with proper steps.	10

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Q 3 (b)	Explain following terms 1.Multivalued attribute 2.Partial participation 3.Total participation 4. Generalization 5. Candidate key	10
Q4 (a)	For a given database, Employee(EID, Name, Address, Country) Works(EID, Dept_ID, Salary) Manager(EID, Manager_Name) Dept(Dept_ID, Dept_name, City) Write SQL queries for, (i)Modify the database so that 'JHON' now lives in USA. (ii)Give all employees of department "R-DIGITAL" a 10% raise in Salary. (iii)Display exact number of cites present in Dept table. (iv)List the addresses of all employees who are Manager. (v)Display name of all employees starting with letter 'A'.	10
Q4 (b)	What is view in SQL? Give complete syntax of creating, updating and deleting view in SQL with example. What are the difficulties when we try to update view? OR Explain all Aggregate functions with example for Order by, Group by and Having clause.(Give at least five queries).	10
Q5 (a)	Compare Shadow Paging Recovery scheme in terms of ease of implementation and overhead cost. OR Explain Time Stamps based Concurrency Control Protocol with example.	10
Q5 (b)	What do you mean by serializable schedule? How would you test whether given schedule S is conflict serializable or not.	10